

Building Foundation Model in Consortium Mode

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BharatGen

GenAI for Bharat, by Bharat



Funded by



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OUR MODEL SUITE

Text. Voice. Documents. Across Indian languages

TEXT · LLM

Param Series

Multilingual · Reasoning, code, math, and tool use · 22 Indian languages + English · Multiturn, instruction-tuned

SPEECH · ASR

Shrutam Series

LLM-powered · MoE projector · 12 Indian languages released · Realtime transcription · Code-switching

VISION · DOCUMENT VLM

Patram Series

Vision transformer · Document QA and information extraction · Multi-domain document support · Table/chart understanding

SPEECH · TTS

Sooktam Series

Cross-lingual voice cloning · CLS · 11 Indian languages + English released · Zero-shot voice generation · Natural, human-sounding output

Talk Overview

What it takes to build Foundational AI?

The Model Development Life Cycle

Glimpses on Training BharatGen's Param 17B Foundation Model – A Case Study!

The GAP -> The Dilemma -> The Proposal -> Federated Learning – The Concept

The Consortium Model

Core Pillars of a Industrial Foundation Model

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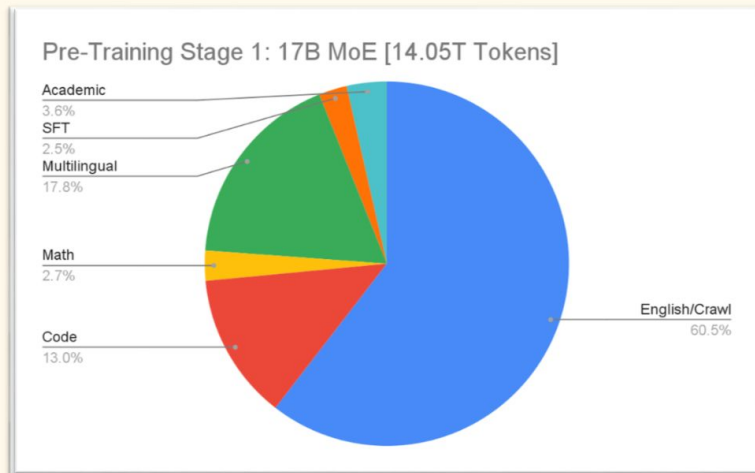


Training Dataset details for the Param 17B Foundation Model

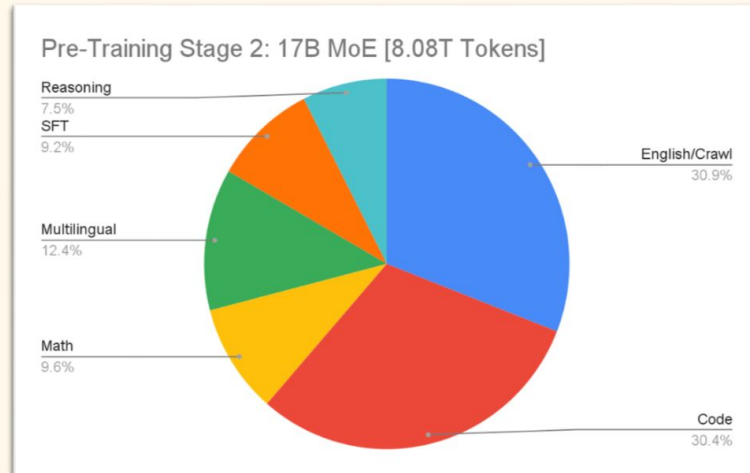
PARAM 2 17B MoE - DATA DETAILS

Dataset Preparation Details: Multi-Stage Pretraining on 22.13 Trillion Tokens

Pre-Training Stage 1 [14.05T]



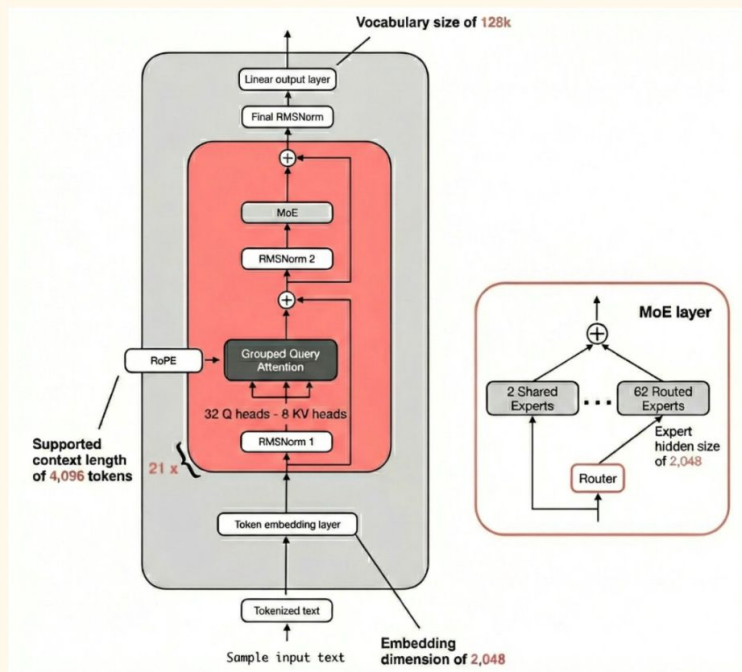
Pre-Training Stage 2 [8.08T]



BharatGen's Flagship Model: Param-17B Mixture of Experts LLM

Param-2-17B-MOE-A2.4B: High-Efficiency Multilingual Powerhouse

State-of-the-art Mixture-of-Experts (MoE) model with 17B total parameters (2.4B active per token), combining massive knowledge capacity with fast, efficient inference, trained on ~22T tokens across 64 experts. Supports 22 Indian languages with strong code, math, and tool-use reasoning, optimized for culturally grounded, region-specific tasks and inclusive representation.



Languages supported		
• Assamese	• Kannada	• Punjabi
• Bengali	• Kashmiri	• Sanskrit
• Bodo	• Konkani	• Santali
• Dogri	• Maithili	• Sindhi
• Gujarati	• Malayalam	• Tamil
• Hindi*	• Manipuri	• Telugu
• Marathi	• Nepali	• Urdu
• Odia (Oriya)		

Now available for access through:

- **AIKosh:** [Link](#)
- **Hugging Face:** [Link](#)

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A Large Language Model, Don't Know Your Factory



What industry gets from frontier APIs today

- Customer & dealer support copilots — the "lowest-hanging fruit"
- Documentation drafting & review, SDS retrieval
- RAG over manuals; part identification from photos
- Sales funnel & after-sales automation



What they admit doesn't work

- ~X% accuracy on documentation review — human refinement needed
- Machine diagnostics & error-code interpretation unreliable
- Weak numerical / engineering reasoning in real workflows
- Fix on offer: Forward Deployed Engineers — humans bolted around the API. The model itself never learns your domain.

These models were trained on the open web. **Your process dynamics, failure modes and SOPs are not on the web — and never will be.**

The Dilemma

Secret Sauce vs. Data Horizon



TRUTH 1

Your expertise is your secret sauce

- It cannot cross your company boundary. Ever.
- Any proposal that starts with “share your data” is dead on arrival — correctly so.
- So industrial knowledge stays fragmented, plant by plant.



TRUTH 2

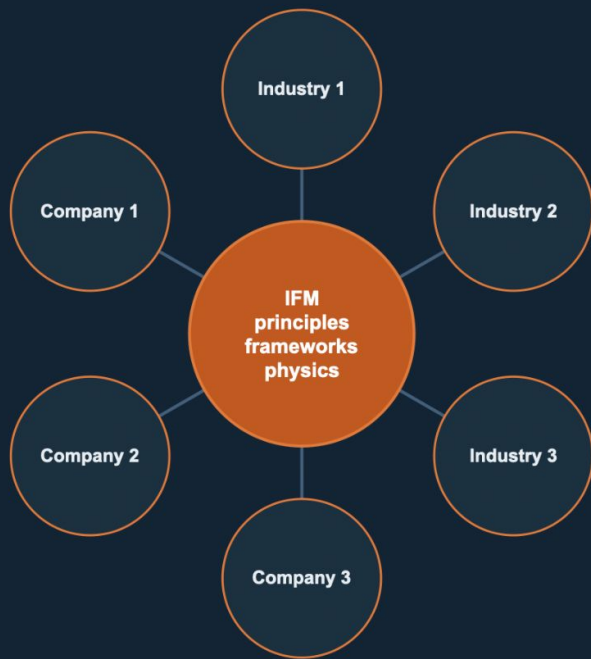
A model trained only inside your boundary plateaus

- Realism is multivariate — one plant sees a sliver of process regimes, failure modes, disturbances.
- Your model overfits to your own history.
- The performance jump that justifies the investment needs breadth no single company has.

Solo, you train an expert in your past. **Together, we train an expert in your industry's future.**

The Proposal

One Model, N Sovereign Nodes



BYOE

Bring Your Own Expertise — not your data.

- N sovereign industry nodes — air-gapped, each the expert at the start
- Data stays on-premise; only latent updates & distilled representations travel
- The central node abstracts principles & frameworks — owned by no one, shared by all
- CI-AIM / IIT Bombay: the neutral, trusted aggregator
- End state: the central model is the smartest of the lot — and every node takes it home

CONFIDENTIAL DATA NEVER LEAVES THE NODE

LATENT KNOWLEDGE TRAVELS

Federated Learning – The Concept

THE CONCEPT · FEDERATED LEARNING

Federated Learning, In One Picture

1 BROADCAST MODEL

2 TRAIN LOCALLY

3 RETURN UPDATES ONLY

4 AGGREGATE → REPEAT



Data never moves. Knowledge compounds. Round after round, the shared model becomes greater than the sum of its parts.

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This Is Already Happening



BharatGen — proven at country scale

- India's first government-funded sovereign AI — DST & IndiaAI Mission (MeitY)
- Consortium of India's top institutes, anchored at IIT Bombay
- 22-trillion-token training runs on national compute
- Sovereign AI works when institutions align



Project Tapestry — federated at global scale

- AI Alliance initiative, led by Yann LeCun
- 30+ organisations: BharatGen, MBZUAI, EPFL, Monash, IBM
- One shared base model — “weights travel, data stays”
- Anti-capture by architecture, not by policy promises

CI-AIM is the industrial chapter of the same story. Nations proved the consortium model. Industries are next.

The Path to Frontier Sovereign AI

A global initiative hosted by the AI Alliance to build frontier-capable foundation models through distributed collaboration — while every participant keeps sovereign control over its data and model derivatives.

Led by Yann LeCun, Chief Science Advisor of the AI Alliance, with IBM, Aitomatic and partners across Europe, Asia and the Middle East.

“There is no meaningful sovereignty without frontier performance.”

— AI Alliance, Project Tapestry announcement

30+

organizations at the May 2026 Paris workshop — BharatGen, MBZUAI, EPFL, IBM, NVIDIA, AMD & more

HOW IT WORKS



One shared base model

Trained jointly by the consortium as a common frontier foundation



Sovereign local training

Each node trains on its own data, under its own controls and jurisdiction



Weights travel, data stays

Participants return weight updates for aggregation — never raw data



Anti-capture by design

Sovereignty enforced by the architecture itself, not by policy promises

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Industrial Foundation Model

Three Pillars of Training an IFM



Off-policy learning — *IFM-Base*

Learning from demonstrations

- Diverse simulations, digital-twin rollouts, augmented industrial data (time series, tables, P&IDs → model-readable form)
- The model distills knowledge — starts reasoning from first principles
- This layer is OPEN & FEDERATED — foundational learning happens in the open format



On-policy learning — *Derivative*

Learning from practice

- The model exercises what it learnt — takes actions in your workflows, calls your tools
- Learns from verifiable outcomes in your processes
- LOCALISED — runs on-premise, inside your air-gapped node



Evaluation & benchmarking

The steering signal

- A realistic benchmark suite lives on EVERY sovereign node
- Benchmarks decide everything: how far back in the pipeline to go, whether the model generalises
- Realism is multivariate → benchmarking itself must be federated

Demonstrations teach knowledge. Practice teaches judgment. Benchmarks tell the truth.

Thank You

The Whole is greater than the sum of its part!